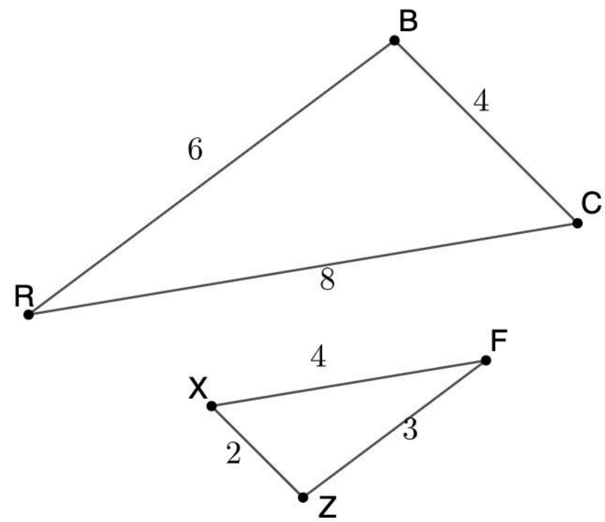
Geometry
Unit 6
Lesson 8 Practice

Name Answer Key________
Date____________________
Period__________________

$\triangle RBC$ and $\triangle XFZ$ are shown.



1. Complete the table.

Measurements of $\triangle RBC$	**Measurements of $\triangle XFZ$**	**Ratio**
$RB = 6$	$ZF = 3$	$\frac{RB}{ZF} = \frac{6}{3} = 2$
$BC = 4$	$XZ = 2$	$\frac{BC}{XZ} = \frac{4}{2} = 2$
$CR = 8$	$XF = 4$	$\frac{CR}{XF} = \frac{8}{4} = 2$

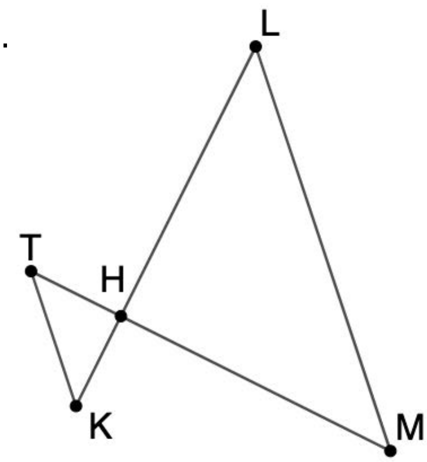
2. What similarity theorem can be applied to show that the triangles are similar?
Side-Side-Side Similarity

3. Write the similarity statement.
$\triangle RBC \sim \triangle FZX$

4. Complete each.

$$\frac{RC}{CB} = \frac{FX}{XZ} \qquad \frac{XF}{FZ} = \frac{CR}{RB} \qquad \frac{BC}{RB} = \frac{ZX}{FZ}$$

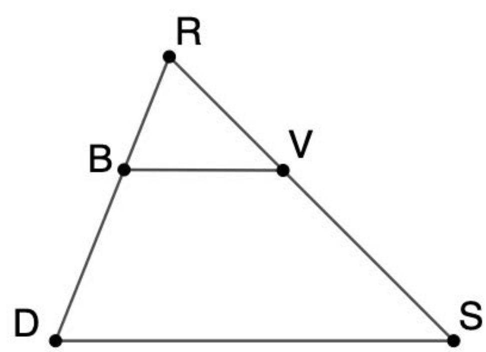
$\triangle TKH$ and $\triangle HLM$ are shown where $\overline{TK} \parallel \overline{LM}$.



Complete the two-column proof.

	Statement	Reason
	$\overline{TK} \parallel \overline{LM}$	Given
5.	$\angle HLM \cong \angle HKT$ and $\angle HML \cong \angle HTK$	If two parallel lines are intersected by a transversal, alternate interior angles are congruent.
6.	$\triangle TKH \sim \triangle MLH$	AA ~

$\triangle DRS$ and $\triangle RVB$ are shown.



Given: $RB = \frac{2}{5}RD$ and $RV = \frac{2}{5}RS$

Prove: $\triangle DRS \sim \triangle BRV$

	Statement	Reason
	$RB = \frac{2}{5}RD$ and $RV = \frac{2}{5}RS$	Given
7.	$\frac{RB}{RD} = \frac{2}{5}$ and $\frac{RV}{RS} = \frac{2}{5}$	Division Property of Equality
8.	$\frac{RB}{RD} = \frac{RV}{RS}$	Transitive Property of Equality
9.	$\angle DRS \cong \angle BRV$	Reflexive Property of Congruence
10.	$\triangle DRS \sim \triangle BRV$	SAS ~